Data Flow Diagram

Date	03 October 2026
Team ID	2
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	2 Marks

Data Flow Diagram

The diagram shows how data moves between FitBuddy's external entities, processes and data stores (Level 1 DFD).

DFD Symbol Legend

Symbol	Name	Used in FitBuddy for
Rounded box (blue)	External Entity	User (fitness seeker), Admin / Coach, Google Gemini API
Rectangle with numbered header	Process	1.0 Generate plan, 2.0 Generate tip, 3.0 Revise plan, 4.0 Admin management
Rectangle (green fill, no number)	Data Store	D1 users, D2 plans (versioned)
Labeled arrow	Data Flow	Profile data, prompts, plan JSON, feedback, dashboard data

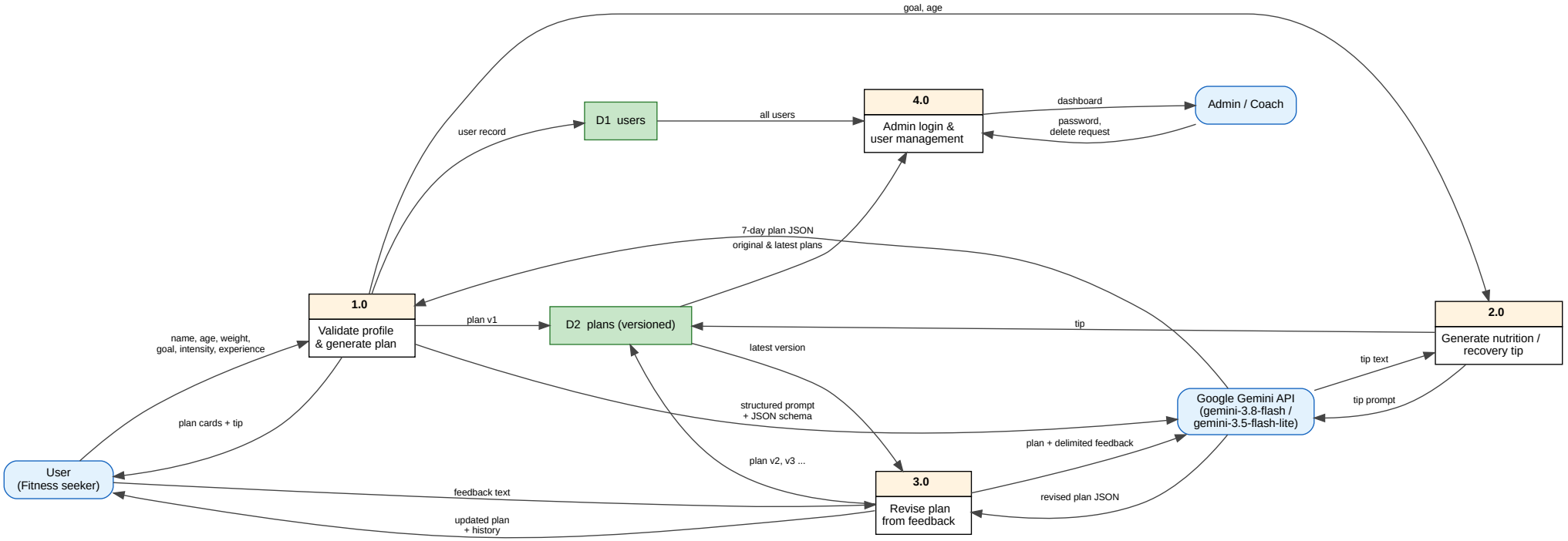


Figure 1: FitBuddy Level-1 Data Flow Diagram

Process descriptions

Process	Inputs	Outputs	Data stores
1.0 Validate profile & generate plan	Name, age, weight, goal, intensity, experience (validated by Pydantic)	7-day plan JSON (validated: 7 days, ≥1 rest day)	Writes D1 user, D2 plan v1 — only after a valid plan is returned
2.0 Generate nutrition / recovery tip	Goal, age	2–3 sentence tip (built-in tip if the model fails)	Writes tip with the plan version in D2
3.0 Revise plan from feedback	Feedback text (≤500 chars, wrapped in delimiters) + latest plan	Revised plan JSON	Reads latest version from D2, writes new version (v2, v3 …)
4.0 Admin login & user management	Admin password, delete request	Dashboard of all users with original vs latest plan	Reads D1, D2; deletes user + plans (cascade)

Note: If any Gemini call fails, process 1.0 / 3.0 returns a friendly error and nothing is written to the data stores, so an error message is never saved as a plan.